



01

LECTURE

BIOSTATISTICS AND EPIDEMIOLOGY

MT 2, First Semester LECTURE 01 - Cleofin Zyrielle Uy, RMT

INTRODUCTION TO BIOSTATISTICS AND VARIABLES

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LEARNING OBJECTIVES

- Define biostatistics and describe its role in analyzing biological data, guiding research, and informing healthcare decisions.
- Summarize the historical development of biostatistics, highlighting key milestones and contributions to modern medical research.
- Define and distinguish between different types of variables in the context of biostatistical research.

I. INTRODUCTION TO BIOSTATISTICS

What is Biostatistics?

Bio = life

Statistics = working with numbers or data

In Medical Technology, we deal with data involving:

- Patients
- Diseases
- Bacteria
- Blood results
- Genes
- Vaccines
- Other biological and medical information

Therefore, **biostatistics** is the application of statistics to biology, medicine, and health-related fields. It is about using data to **make decisions and draw interpretations**.

II. DEFINITION OF STATISTICS

King's Definition

- The science of statistics is the method of judging collective, natural, or social phenomena.
- **Keyword:** Collective
- Statistics focuses on **groups or patterns within a population**, rather than just one individual.
- Example:
 - Instead of asking: "Does Mary have diabetes?"
 - Statistics would ask: "Among 1,000 Filipinos aged 40–60, how many have diabetes?"

Lovitt's Definition

- Statistics is the science which deals with collection, classification, description, and comparison of phenomena.
- **Keywords:** Process of handling data.
 - **COLLECT** → **CLASSIFY** → **DESCRIBE** → **COMPARE**
- Example:
 - Suppose a researcher studies the blood glucose levels of 100 patients.
 - Collect - Measure the patients' blood glucose.
 - Classify - Group them as normal, prediabetic, or diabetic.
 - Describe - Identify patterns, such as how many patients have the highest glucose levels.
 - Compare - Compare glucose levels between males and females or between different age groups.

Croxton and Cowden's Definition

- Statistics is the science which deals with the collection, analysis, and interpretation of numerical data.
- Keyword:** Interpretation
- Example: A researcher compares Drug A and Drug B for lowering blood glucose. After collecting the data, the researcher analyzes the results and determines that: Drug A works significantly better than Drug B.

SUMMARY		
King	Lovitt	Croxton & Crowden
Collective - Groups - Population	Process - Collect - Classify - Describe - Compare	Analysis & interpretation

Table 1. Summary of the Definitions of Statistics

III. HISTORY OF BIOSTATISTICS

- Biostatistics is defined as statistics arising out of biological sciences, particularly from the fields of medicine and public health.
- The methods used in dealing with statistics in the fields of medicine, biology, and public health for **planning, conducting, and analyzing data** which arise in investigations of these branches.

HISTORY OF STATISTICS IN MEDICAL RESEARCH	
1937	- Austin Bradford Hill, a pioneering figure in medical statistics, published 15 influential articles on statistical methods. - Introduced proper statistical methods to medicine; Medicine is evidence-based.

1948	- Publication of a randomized controlled trial (RCT) on the use of streptomycin for treating pulmonary tuberculosis. - First randomized clinical trial Gold standard: Clinical trial
1952–1982	The field of medical statistics experienced exponential growth, with an eightfold increase in publications related to statistics in medicine by 1982.
1992	A landmark paper by Dunn was published in the <i>Journal of Physiology</i>, marking one of the earliest comprehensive applications of statistical methods to physiological research.

Table 2. History of Statistics in Medical Research

NOTABLE STATISTICIANS

Douglas Altman

- Pioneered medical statistics, focusing on statistical errors and improving clinical research reporting.
 - He improved medical research quality
 - Improve reducing statistical errors

Ronald Fisher

- Revolutionized statistics with ANOVA, maximum likelihood estimation, and population genetics.
 - Father of Experimental Statistics
 - Without him, clinical trial will not exist

Carl Gauss

- Developed the method of least squares and the Gaussian distribution.

- Developed the bell-shaped, Gaussian distribution

Karl Pearson

- Founded modern statistics and created the Pearson correlation coefficient.
 - Measures the relationship
 - Example: If you study longer, you'll get a good score

C. R. Rao

- Advanced estimation theory and multivariate analysis with the Cramer-Rao bound and Rao's score test.

IV. SIGNIFICANCE OF BIOSTATISTICS

- Biostatistics is the science that helps in managing medical uncertainties. (Indrayan, 2021).
- Sources of medical uncertainty:**
 - Intrinsic factors
 - Biological factors
 - Environmental factors
 - Sampling factors
 - For example: Why the COVID-19 treatment can cure this person and not the other.
 - Natural variation
 - Differences among methods and techniques
 - Differences among observers
 - Differences among instruments
 - Measurement/assessment errors
 - Errors in measurements
 - Errors in assessment
 - Errors in knowledge
 - Incomplete knowledge
- Medicine is becoming increasingly quantitative and evidence-based

- The planning, conduct and interpretation of much of medical research are becoming increasingly reliant on statistical methodology.
- Statistics pervades the medical literature.

EXTRA NOTE!

Medicine deals with uncertainty. Biostatistics helps researchers and healthcare professionals quantify and manage that uncertainty instead of relying purely on assumptions.

BIostatISTICS IN CLINICAL MEDICINE

- Systematically documenting and analyzing their medical history of diseases, enabling the identification of patterns, risk factors, and trends.
- Essential in the design and execution of clinical studies to produce reliable and valid results.
- Compare and evaluate the efficacy, safety, and cost-effectiveness of different medical procedures and treatments.
- Establishes the statistical criteria for defining what is considered "normal" and abnormal in clinical measure.

BIostatISTICS IN PATIENT CARE

- Identify and quantify the uncertainties inherent in diagnosis, treatment, and prognosis.
 - To know which test is used for a certain diseases
- Provides methods for combining statistical probabilities with clinical expertise and patient values.
- Through techniques such as sensitivity, specificity, and predictive values; biostatistics helps clinicians select the most appropriate diagnostic tests.
- Aids in the development of scoring systems and expert systems that systematically reduce uncertainties in clinical practice.

BIostatISTICS IN PREVENTIVE MEDICINE

- Quantifies the prevalence and incidence of health problems within a community, offering a clear picture of the burden of diseases.
- Identify and quantify the underlying risk factors and determinants of ill-health in a population, guiding the development of targeted prevention strategies and public health policies.
- Assessing the effectiveness of health programs implemented in the community.
- Provides the data needed to support the development and advocacy of health legislation aimed at preventing disease, promoting health, and improving the overall well-being of the population.

BIostatISTICS IN HEALTH PLANNING AND EVALUATION

- Enables the accurate collection, summarization, and interpretation of health data; ensuring that health situation analyses are both valid and reliable.
- Crucial in objectively assessing the outcomes of health programs by measuring their impact, identifying successes, and pinpointing areas of failure.

BIostatISTICS IN MEDICAL RESEARCH

- Essential in creating robust research designs that minimize the impact of biases and uncertainties.
- Biostatistics plays a crucial role in evaluating the reliability (consistency) and validity (accuracy) of the tools and instruments used to collect data in medical research.
- Fundamental in the correct analysis of data, ensuring that appropriate statistical techniques are used to interpret the results accurately.

V. BASIC CONCEPTS OF STATISTICS: PROCESS

- The science of conducting studies from data to:
 - Collect
 - Organize
 - Summarize
 - Analyze
 - Present
 - Interpret
 - Draw conclusions

4 MAJOR ACTIVITIES	
Collecting Data	Sample, survey, observation, simulation
Characterizing Data	Organize, classify, count, summarize
Presenting Data	Tables, charts, statements
Interpreting Data	Infer, conclude, specify confidence

Table 3. 4 Major Activities

VI. BASIC CONCEPTS OF STATISTICS: DATA

- **Data** is a set of values of one or more variables recorded on one or more observational units.
 - Data are not only values, it can also be description, and perspective of people
- Sources of data:
 - Routinely kept records
 - Surveys
 - Experiments
 - External sources

HISTORY OF STATISTICS IN MEDICAL RESEARCH	
Primary Data	• Data collected by YOU (the researcher). ◦ Observation

	- Questionnaire - Interviews - Surveys
Secondary Data	- Data collected by SOMEBODY ELSE then used by the researcher. - Medical records - Registry - Census

Table 4. History of Statistics in Medical Research

Dependent Variable (DV)

- Variables that are **measured** in a study to see if they are affected by the independent variables.
- DV = effect or output**

Moderator Variable

- Variables that may **influence** the relationship (strength or direction) between the independent and dependent variables.

Mediator Variable

- Variables that explain how or why the independent variable (IV) affects the dependent variable (DV).
- IV → MEDIATOR → DV

APPLICATIONS	
Number of Days of Blood Storage and RBC Viability	
INDEPENDENT VARIABLE	Number of days of blood storage
DEPENDENT VARIABLE	RBC viability
MODERATOR VARIABLE	Storage temperature Influences the strength of this relationship because RBCs may respond differently to storage duration depending on the temperature at which the blood is stored.
Medication Dosage and Blood Pressure	
INDEPENDENT VARIABLE	Medication dosage

DEPENDENT VARIABLE	Blood pressure
MODERATOR VARIABLE	Physical activity Physically active patients might respond differently to medication dosage. Active patients may require a lower dosage.

Table 5. Applications of the Types of Variables

VII. BASIC CONCEPTS OF STATISTICS: VARIABLES

- A **variable**, as applied in research, is the subject or main focus of your study
- It involves all aspects or characteristics related to the subject or problem that can be altered or constant in nature.
- A characteristic or attribute of persons or objects which assumes different values or labels.
- Examples:
 - Age
 - Height
 - Sex
 - Blood type
 - Blood pressure
 - Weight
 - Smoking status
- Anything that can vary can potentially be a variable.

TYPES OF VARIABLES

Independent Variable (IV)

- Variables that are **manipulated or changed** in an experiment to examine their effects on the dependent variable.
- IV = the cause or input**

VIII. SCALES OF MEASUREMENT

- There are four scales of measurement with a usual progression of:
 - Nominal
 - Ordinal
 - Interval
 - Ratio

Nominal Scale

- The most basic level of measurement.
- It categorizes variables without providing value or order.
- Type of Data:** Qualitative
- Examples:** Gender, Marital status, Categories
- Key characteristic:**
 - NO ORDER OR RANKING
 - You cannot say that one category is "higher" or "lower" than another.

Ordinal Scale

- Represents variables with a meaningful rank or order, but the intervals between categories are not necessarily equal or known.
- Type of Data:** Qualitative
- Examples:** Socioeconomic status, Educational attainment

- You could arrange education levels:
Elementary → High School → College → Master's
 - There is an order, but the "distance" between the categories isn't necessarily equal.
- You could rank socioeconomic status, but the difference between categories isn't a fixed numerical interval.
- **Key characteristic:**
 - HAS ORDER OR RANK, BUT NO EQUAL INTERVAL

Interval Scale

- Orders and measures variables with their exact differences between them however it **lacks a true zero point**.
- **Type of Data:** Quantitative
- **Examples:** Temperature, IQ scores
 - The difference between 20°C - 30°C, is the same as 30°C - 40°C. However, 0°C does not mean the complete absence of temperature.
- **Key characteristic:**
 - EQUAL INTERVALS + NO TRUE ZERO
 - The zero in an interval scale **does not** represent a true absence of the quantity.

Ratio Scale

- The highest level of measurement.
- It has all the characteristics of an interval scale but also has a true zero point.
- **Type of Data:** Quantitative
- **Examples:** Weight, Income, Distance
 - 0kg = absence of weight
 - ₦0 = absence of money
 - 0km = absence of distance traveled
- **Key characteristic:**
 - EQUAL INTERVALS + TRUE ZERO
 - A true zero means that zero represents the complete absence of the quantity being measured.

SUMMARY				
Scale	Order	Equal Intervals	True Zero	Data
Nominal	NO	NO	NO	QUALI
Ordinal	YES	NO	NO	QUALI
Interval	YES	YES	NO	QUANTI
Ratio	YES	YES	YES	QUANTI

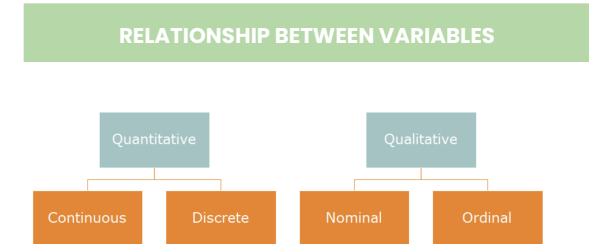
Table 6. Summary of the Scales of Measurement

CONVERSION OF SCALES OF MEASUREMENT

- **Direction of Conversion:**
 - Ratio → Interval → Ordinal → Nominal
- Higher levels of measurement can be converted into a lower level.
 - We cannot convert Nominal to Ratio since we cannot determine the original numerical values from the nominal categories alone.

APPLICATIONS							
Ratio to Ordinal							
RATIO: Numerical Value ● 50 mg/dL ● 121 mg/dL	ORDINAL: Rank or Order ● Low < High <table> <tr> <td>Glucose Level</td><td>Classification</td></tr> <tr> <td>50 mg/dL</td><td>Low</td></tr> <tr> <td>121 mg/dL</td><td>High</td></tr> </table>	Glucose Level	Classification	50 mg/dL	Low	121 mg/dL	High
Glucose Level	Classification						
50 mg/dL	Low						
121 mg/dL	High						

Table 7. Application of the Scales of Measurement



Discrete

- Values that are countable, distinct, and separate.
- Usually **whole numbers**.
- Cannot have values in between.
- **Example:** Number of doors in a latrine
 - 1, 2, or 3

Continuous

- Values that can be measured.
- Can have any value within a range, **including decimals and fractions**.
- **Example:** Time in minutes
 - 5 min, 5.5 min, 5.25 min, etc.

IX. CLINIMETRICS

- Clinimetrics is the science in which qualities are converted into meaningful quantities using a **scoring system**.
- It allows:
 - Standardization of clinical assessments
 - Decision-making
 - Research
 - Evaluation

EXAMPLES OF CLINIMETRICS

APGAR Score

- Used for **neonatal prognosis**
- Based on:
 - A – Appearance
 - P – Pulse
 - G – Grimace
 - A – Activity
 - R – Respiration

Smoking Index

- Based on factors such as:
 - Number of cigarettes
 - Duration of smoking
 - Whether filtered or unfiltered
 - Type of smoking:
 - Pipe
 - Cigar
 - Etc.

APACHE Score

- **APACHE** = Acute Physiology and Chronic Health Evaluation
- Used to quantify the severity of a patient's condition.

X. REFERENCES

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